

FULL-LENGTH ORIGINAL RESEARCH

Clinical comparability of the new antiepileptic drugs in refractory partial epilepsy: A systematic review and meta-analysis

*†João Costa, *Filipa Fareleira, *Raquel Ascensão, *†Margarida Borges, †Cristina Sampaio, and *António Vaz-Carneiro

*Center for Evidence-Based Medicine, University of Lisbon School of Medicine, Lisbon, Portugal; and
†Laboratory of Clinical Pharmacology and Therapeutics, Faculty of Medicine, University of Lisbon, Lisbon, Portugal

SUMMARY

Purpose: Evaluate the clinical comparability of new antiepileptic drugs (AEDs) in partial refractory epilepsy.

Methods: Systematic review of randomized trials (RCTs) comparing a new AED (add-on treatment) with placebo or another AED. Primary outcomes: responder ($\geq 50\%$ seizure reduction) and withdrawal (tolerability) rates. Pooled estimates of odds ratios (ORs) and number needed treat/harm (NNT/NNH) taking into account baseline risk were derived by random-effects meta-analysis. Adjusted frequentist indirect comparisons between AEDs were estimated.

Key Findings: Sixty-two placebo-controlled (12,902 patients) and eight head-to-head RCTs (1,370 patients) were included. Pooled ORs for responder and withdrawal rates (vs. placebo) were 3.00 [95% confidence interval (CI) 2.63–3.41] and 1.48 (1.30–1.68), respectively. Indirect comparisons of responder rate based on relative measurements of treatment effect (ORs) favored topiramate (1.52; 1.06–2.20) in comparison to

all other AEDs, whereas gabapentin (0.67; 0.46–0.97) and lacosamide (0.66; 0.48–0.92) were less efficacious, without significant heterogeneity. When analyses were based on absolute estimates (NNTs), topiramate and levetiracetam were more efficacious, whereas gabapentin and tiagabine were less efficacious. Withdrawal rate was higher with oxcarbazepine (OR 1.60; 1.12–2.29) and topiramate (OR 1.68; 1.07–2.63), and lower with gabapentin (OR 0.65; 0.42–1.00) and levetiracetam (OR 0.62; 0.43–0.89).

Significance: The differences found are of relatively small magnitude to allow a definitive conclusion about which new AED(s) has superior effectiveness. This uncertainty probably reflects the limitations of conclusions based on indirect evidence. The process of pharmacologic clinical decision making in partial refractory epilepsy probably depends more on other aspects, such as individual patient characteristics and pharmacoeconomics, than on available controlled randomized evidence.

KEY WORDS: Epilepsies, Partial, Anticonvulsants, Meta-analysis, Comparative effectiveness research.

Over the past 15 years, several new antiepileptic drugs (AEDs) have been introduced with the global aim of providing a better control of the seizures and a more favorable safety and tolerability profile over the so-called old AEDs. To obtain regulatory approval for marketing use, the target indication is usually add-on treatment for partial refractory epilepsy. Therefore, all new AEDs compared themselves with placebo in several randomized clinical trials (RCTs). Cochrane systematic reviews (Castillo et al., 2000; Marson et al., 2000; Chaisewikul et al., 2001; Ramaratnam et al., 2001; Pereira et al., 2002; Chadwick & Marson, 2005; Jette

et al., 2008; Lozsadi et al., 2008) have shown without exceptions that, in comparison to placebo, all new AEDs provide a better control of frequency of seizures in this epileptic population. Therefore, both clinicians and decision makers have to choose among these competing interventions. However, due to few head-to-head trials available, little is known about their relative effectiveness. This systematic review aimed to evaluate how the new AEDs compare in the treatment of patients with partial refractory epilepsy.

METHODS

Studies

Criteria for inclusion of trials: (1) adequate allocation concealment, blinding of interventions and assessment; (2) patients older than 2 years of age with a clinical diagnosis

Accepted February 1, 2011; Early View publication April 19, 2011.

Address correspondence to João Costa, CEMBE, Faculty of Medicine, Av Prof Egas Moniz, 1649-028 Lisboa, Portugal. E-mail: joaocosta@sapo.pt

Wiley Periodicals, Inc.

© 2011 International League Against Epilepsy

of drug-resistant partial epilepsy (partial onset seizures—i.e., experiencing simple partial, complex partial, or secondarily generalized tonic-clonic seizures—that have failed to respond to monotherapy with an AED); (3) sample size ≥ 10 patients per group; (4) one of the following new AEDs in the intervention arm used as add-on treatment [oxcarbazepine (OXC), lamotrigine (LTG), topiramate (TPM), gabapentin (GBP), pregabalin (PGB), levetiracetam (LEV), tiagabine (TGB), zonisamide (ZNS), eslicarbazepine (EBZ), or lacosamide (LCS)]; (5) matched placebo or other new AED as control; (6) minimum follow-up of 8 weeks for treatment period (excluding titration), as this is thought to represent the minimum period over which changes in seizure frequency can be determined; and (7) provision for or allowing calculation of results for the primary outcomes. We consider both parallel and crossover trials. Data analysis of crossover studies was based exclusively on the first treatment period. Studies evaluating only special groups (e.g., institutionalized, cognitively impaired, pregnant) were excluded.

Outcome measures

Primary outcomes

(1) Responder rate (proportion of patients experiencing a 50% or greater reduction in seizure frequency in the treatment period compared to the prerandomization baseline period), and (2) withdrawal rate (proportion withdrawing from treatment during treatment period). Responder rate is a global measure of efficacy commonly reported in RCTs and can be calculated provided that baseline seizure data were recorded. Withdrawal rates capture global tolerability and effectiveness because in studies of relatively short duration, adverse effects are likely the main reason for withdrawal. This is an outcome to which participants can make a direct contribution.

Secondary outcomes

(1) Proportion of patients that were seizure-free during treatment period, (2) withdrawal rate due to adverse effects, and (3) proportion of patients experiencing any of the following commonest adverse effects: ataxia, dizziness, fatigue, headache, nausea, and somnolence.

Search strategy, study selection, and data extraction

Studies were identified through electronic searching of bibliographic databases and reference lists, up to October 2009. Selection of studies and data extraction was done independently and cross-checked for accuracy (Data S1).

Statistical analyses

We performed three analyses. First, we compared each AED with placebo using random-effects meta-analysis weighted by the inverse-variance method to estimate pooled odds ratios (ORs) and 95% confidence intervals

(CIs). Based on these estimates we calculated the number needed to treat (NNT) or to harm (NNH) and 95% CI taking into account the baseline risk (event rate in control group). Subgroup analyses were done according to dose. Heterogeneity was assessed with the I^2 test that measures the percentage of total variation across studies due to heterogeneity (Deeks et al., 2001). Second, we performed adjusted indirect comparisons between each AED and the pooled estimate effect of all other AEDs using the Bucher frequentist method (Bucher et al., 1997) in which the indirect comparison is calculated from the results of their direct comparisons against a common intervention control (placebo) estimated by random-effects meta-analysis. Third, we combined the evidence generated by indirect comparison with evidence from head-to-head RCTs using random effects model for quantitative pooling (DerSimonian & Laird, 1986; Glenny et al. 2005) and determined the discrepancy between direct and indirect estimates (Song et al., 2008) (Data S2).

RESULTS

Description of studies

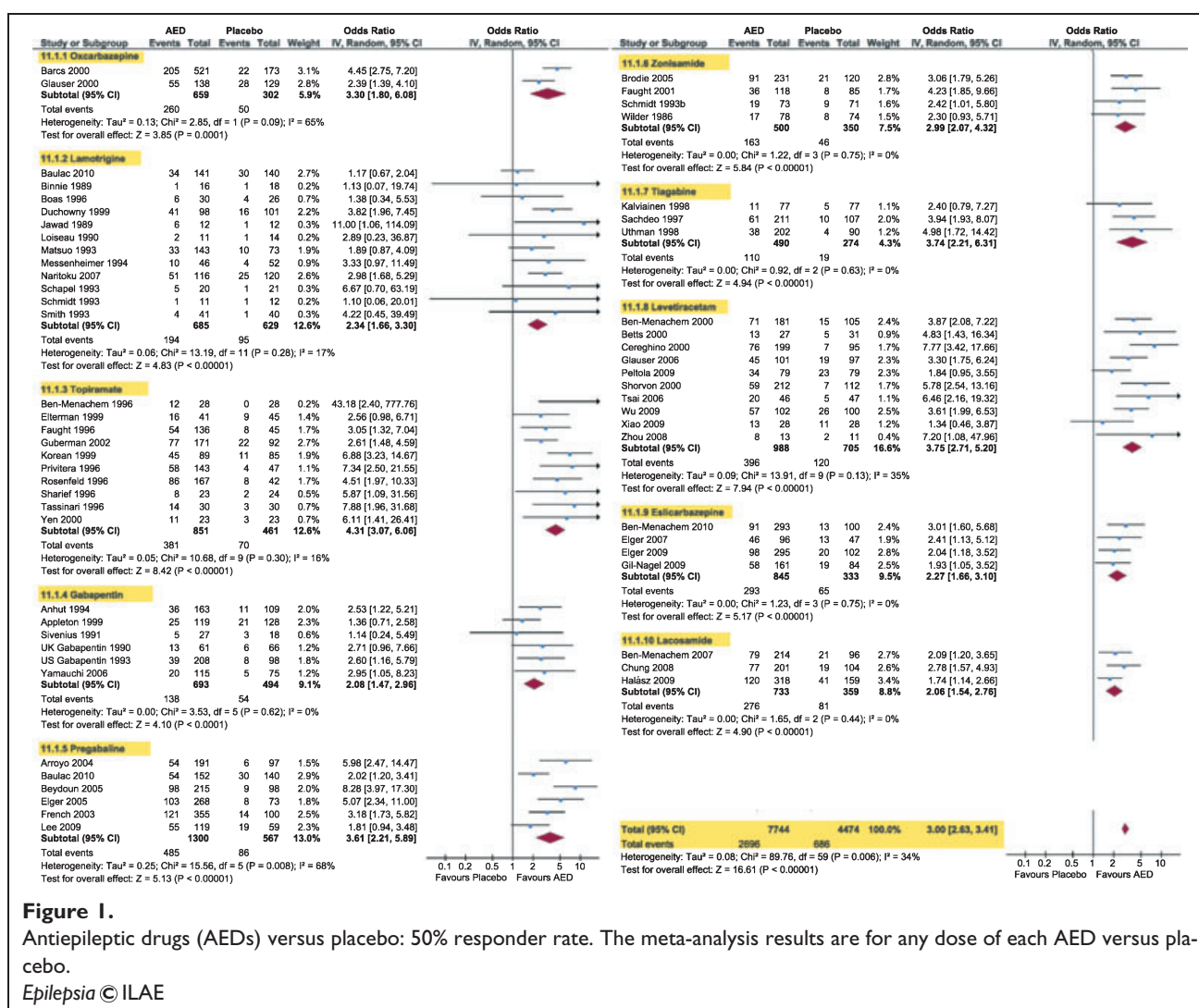
We selected 70 RCTs for analysis: 63 trials compared a new AED with placebo as add-on treatment (12,902 patients) and 8 were head-to-head trials (1,370 patients), 7 of which compared LTG with other new AED. One trial provided both head-to-head and placebo comparisons. Figure S1 shows the results of the search strategy and the reasons for exclusion of trials. The detailed description and main characteristics of the 70 included studies are available at Data S3 and in Tables S1 and S2.

Effects of interventions

AEDs versus placebo

Primary outcomes. Responder rate for each AED (irrespective of the dose) was significantly higher for all AEDs in comparison to placebo, with ORs between 2.08 (GBP) and 4.31 (TPM). Significant heterogeneity was found only for OXC ($I^2 = 65\%$) and PGB ($I^2 = 68\%$). Similar results were obtained for all AEDs, including for OXC and PGB, using the fixed-effects model. Significant differences were found in the dose-subgroup analysis for OXC ($p = 0.02$; $I^2 = 71\%$) and EBZ ($p = 0.03$; $I^2 = 70\%$) suggesting a dose-response relationship for these AEDs. However, the structure of the data within the trials did not allow the performance of a dose-response regression analysis. The results of the efficacy analysis adjusted for baseline risk were also favorable to all AEDs, with NNTs between 8 (LEV) and 20 (TGB). Figure 1 and Table 1 show the detailed results.

Withdrawal rate from any cause was higher with OXC, TPM, PGB, ZNS, TGB, and LCS in comparison to placebo (ORs range, 1.52–2.38; NNHs range, 10–26), but not with LTG, GBP, LEV, and EBZ. Significant heterogeneity was



found only for EBZ ($I^2 = 50\%$). Similar results were obtained for all AEDs using the fixed-effect model. Significant differences were found in the dose-subgroup analysis for OXC ($p < 0.01$; $I^2 = 93\%$), GBP ($p = 0.04$; $I^2 = 63\%$), and ZNS ($p < 0.01$; $I^2 = 88\%$). Figure 2 and Table 1 show the detailed results.

Secondary outcomes. Data on the percentage of patients that were seizure free were available only in 32 of 63 studies. Significant differences favoring AEDs were found for TPM, LEV, and EBZ, without evidence of heterogeneity. Withdrawal rate due to adverse events was higher for LTG, TPM, GBP, PGB, ZNS, EBZ, and LCS, but not for TGB and LEV. However, 15 of 63 studies did not provided data for this outcome. Significant heterogeneity was found for LTG, TGB, and EBZ. The incidence of the six prespecified adverse effects were, in general, higher among all AEDs in comparison to placebo. Table 2 shows the detailed results for these secondary outcomes.

Indirect adjusted comparisons between each AED and all other AEDs

In this analysis, the estimated effect of each individual AED (irrespective of the dose) was compared to the pooled net effect of all other AEDs.

Primary outcomes. Significant differences were found in the analysis of the responder rate based on relative measurements of treatment effect (ORs) favoring TPM (OR 1.52; CI 95% 1.06–2.20) in comparison to all other AEDs. On the other hand, GBP (OR 0.67; CI 95% 0.46–0.97) and LCS (OR 0.66; 95% CI 0.48–0.92) were less efficacious than all other AEDs. For these two AEDs, no significant heterogeneity was found in the meta-analysis. A trend was found for EBZ (OR 0.73; CI 95% 0.52–1.03; $p = 0.07$). However, in the case of EBZ, there were significant differences in the dose-subgroup analysis ($p = 0.03$). A sensitivity analysis including only the results of the most efficacious doses (800 and 1,200 mg daily) showed more favorable results (OR

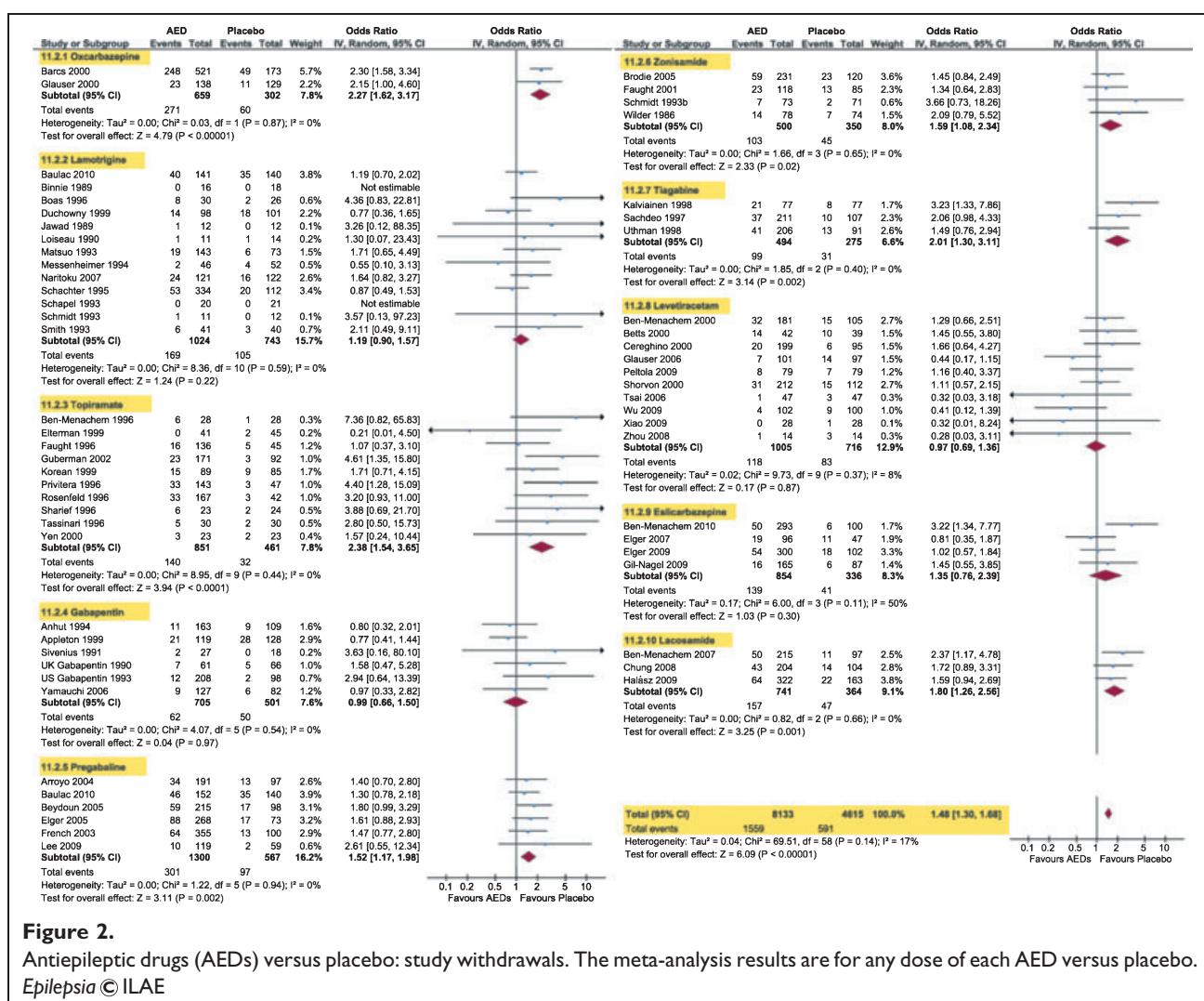
Table 1. Primary outcomes: responder and withdrawal rates—AEDs (any dose) versus placebo

	OXC	LTG	TPM	GBP	PGB	ZNS	TGB	LEV	EBZ	LCS	All AEDs
Responder rate											
RCTs/Pts (n/n)	2/961	12/1,314	10/1,312	6/1,187	6/1,867	4/850	3/764	10/1,693	4/1,178	3/1,092	59/12,218
Frequency (%)	39.5/16.6	28.3/15.1	44.8/15.2	19.9/10.9	37.3/15.2	32.6/13.1	22.4/6.9	40.1/17.0	34.7/19.5	37.7/22.6	34.8/15.3
AED/placebo											
OR (random model)	3.30 (1.80–6.08)	2.34 (1.66–3.30)	4.31 (3.07–6.06)	2.08 (1.47–2.96)	3.61 (2.21–5.89)	2.99 (2.07–4.32)	3.74 (2.21–6.31)	3.75 (2.71–5.20)	2.27 (1.66–3.10)	2.06 (1.54–2.76)	3.00 (2.63–3.41)
OR (fixed model)	3.38 (2.36–4.84)	2.28 (1.71–3.03)	4.15 (3.07–5.60)	2.08 (1.47–2.96)	3.27 (2.49–4.29)	2.99 (2.07–4.32)	3.74 (2.21–6.31)	3.66 (2.85–4.71)	2.27 (1.66–3.10)	2.06 (1.54–2.76)	2.88 (2.61–3.18)
NNT	9 (7–15)	12 (10–18)	9 (8–10)	19 (14–31)	10 (8–13)	12 (10–16)	20 (17–27)	8 (8–10)	10 (8–15)	10 (8–15)	10 (10–11)
Heterogeneity: $p(\chi^2)/I^2$ ^a	0.09/65%	0.28/17%	0.30/16%	0.62/0%	<0.01/68%	0.75/0%	0.63/0%	0.13/35%	0.75/0%	0.44/0%	<0.01/34%
Subgroup analysis (dose): p/I^2 ^b	0.02/71%	0.32/13.5%	0.08/49%	0.20/36%	0.08/59%	0.74/0%	0.23/33%	0.25/26%	0.03/70%	0.15/51%	<0.001/64%
Withdrawal rate											
RCTs/Pts (n/n)	2/961	13/1,767	10/1,312	6/1,206	6/1,867	4/850	3/769	10/1,721	4/1,190	3/1,105	60/12,748
Frequency (%)	41.1/19.9	16.5/14.1	16.5/6.9	8.8/10.0	23.2/17.1	20.6/12.9	20.0/11.3	11.7/11.6	16.3/12.2	21.2/12.9	19.2/12.8
AED/placebo											
OR (random model)	2.27 (1.62–3.17)	1.19 (0.90–1.57)	2.38 (1.54–3.65)	0.99 (0.66–1.50)	1.52 (1.17–1.98)	1.59 (1.08–2.34)	2.01 (1.30–3.11)	0.97 (0.69–1.36)	1.35 (0.76–2.39)	1.80 (1.26–2.56)	1.48 (1.30–1.68)
OR (fixed model)	2.27 (1.62–3.17)	1.19 (0.90–1.57)	2.38 (1.54–3.65)	0.99 (0.66–1.50)	1.52 (1.17–1.98)	1.59 (1.08–2.34)	2.01 (1.30–3.11)	0.98 (0.72–1.36)	1.29 (0.87–1.90)	1.80 (1.26–2.56)	1.49 (1.33–1.66)
NNH	10 (8–15)	NA	26 (20–43)	NA	19 (13–47)	23 (9–119)	19 (14–42)	NA	NA	19 (13–42)	26 (21–38)
Heterogeneity: $p(\chi^2)/I^2$ ^a	0.87/0%	0.59/0%	0.44/0%	0.54/0%	0.94/0%	0.65/0%	0.40/0%	0.37/8%	0.11/50%	0.66/0%	0.14/17%
Subgroup analysis (dose): p/I^2 ^b	<0.01/93%	0.30/17.8%	0.43/0%	0.04/63%	0.08/60%	<0.01/88%	0.09/59%	0.26/25%	0.07/62%	0.35/0%	<0.01/66%

Values in brackets represent the 95% confidence interval.

OXC, oxcarbazepine; LTG, lamotrigine; TPM, topiramate; GBP, gabapentin; PGB, pregabalin; ZNS, zonisamide; TGB, tiagabine; LEV, levetiracetam; EBZ, eslicarbazepine; LCS, lacosamide; AED, antiepileptic drug; RCTs, randomized controlled trials; Pts, patients; OR, odds ratio; NNT, number needed to treat; NNH, number needed to harm; NA, not applicable.

^a A $p(\chi^2 \text{ test})$ value <0.10 and/or $I^2 \geq 50\%$ indicate the presence of significant heterogeneity.^b A $p(z \text{ test})$ value ≤ 0.05 indicates significant differences in the outcome results between subgroups defined according to their daily doses. In the case of All AEDs, the subgroups represent each one of the different AED.



0.84; CI 95% 0.60–1.19; $p = 0.40$). There were also significant differences in the results of the analysis of the responder rate based on absolute estimates (NNTs) adjusted for baseline risk. TPM (NNT 0.83; CI 95% 0.72–0.96) and LEV (NNT 0.78; CI 95% 0.66–0.92) were more efficacious; and GBP (NNT 1.90; CI 95% 1.28–2.80) and TGB (NNT 1.99; CI 95% 1.57–2.52) were less efficacious. In this analysis, a NNT < 1 indicates that fewer patients needed to be treated with that particular AED in comparison to other AEDs for one patient experience a 50% or greater reduction in his seizure frequency. The differences in the results with OR and NNT estimates demonstrate the importance of considering the baseline risk in the analysis. In particular, OR for LCS was significantly different from other AEDs, but not the NNT. This is because the responder rates in the placebo arm were higher in LCS trials (22.6%) than in the trials of the other AEDs. Similar findings were obtained for EBZ, which also had a high responder rate (19.5%) in the placebo group.

There were also significant differences in the withdrawal rate between AEDs. OXC (OR 1.60; CI 95% 1.12–2.29) and TPM (OR 1.68; CI 95% 1.07–2.63) were associated with more withdrawals; and GBP (OR 0.65; CI 95% 0.42–1.00) and LEV (OR 0.62; CI 95% 0.43–0.89) with less withdrawals. The NNHs were not calculated for LTG, GBP, LEV, and EBZ because there were no differences between these AEDs and the placebo. The NNH was 0.33 (CI 95% 0.21–0.52) and 0.92 (CI 95% 0.56–1.52) for OXC and TPM, respectively. In this case, an NNH < 1 indicates that fewer patients needed to be treated with that particular AED in comparison to other AEDs in order for one patient withdraw from the study. Figure 3 shows the detailed results of the indirect adjusted comparisons.

Secondary outcomes. There were no significant differences between AEDs in the proportion of patients that were seizure free. However, the data available for this comparison is sparse. Withdrawal rate due to adverse events was

Table 2. Secondary outcomes: AEDs (any dose) versus placebo

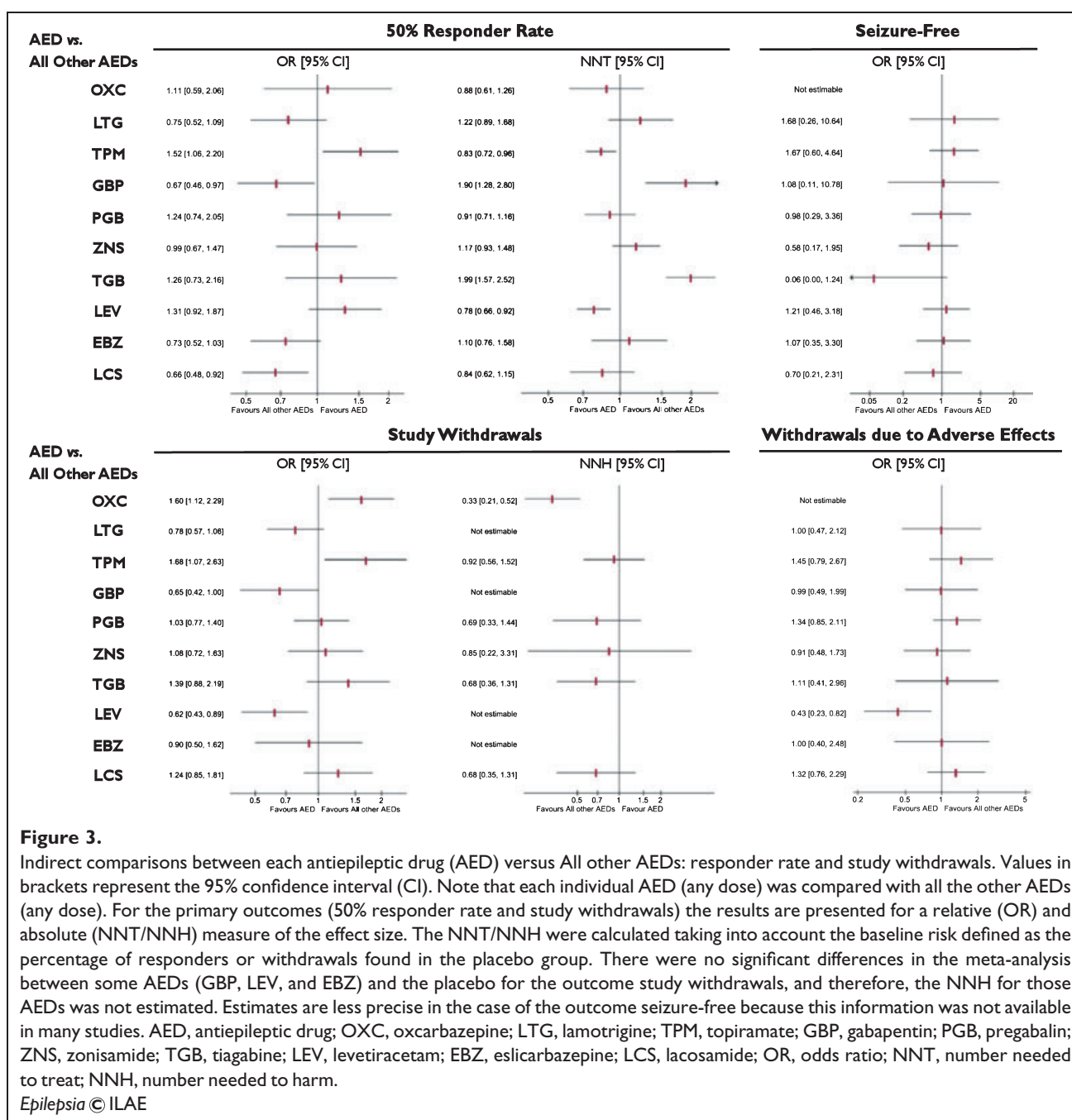
	OXC	LTG	TPM	GBP	PGB	ZNS	TGB	LEV	EBZ	LCS	All AEDs
Seizure-free RCTs/Pts (n/n)	0 3/522	6/839 2.0/0.4	2/437 6.3/0.9	5/1,689 1.9/0.4	2/554 3.4/2.0	1/154 0.0/2.6	10/1,682 6.5/2.1	3/1,033 4.5/1.4	3/968 2.8/0.9	34/7,880 3.8/1.2	
Frequency (%): AED/placebo	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
OR ^a	5.00 (0.82–30.4)	4.73 (1.73–12.9)	3.28 (0.34–32.0)	3.01 (0.94–9.61)	1.87 (0.59–5.88)	0.19 (0.01–4.13)	3.49 ^b (1.49–8.13)	3.24 (1.14–9.26)	2.22 (0.72–6.84)	3.05 (2.09–4.46)	
Heterogeneity: I ² , %	NA	0	0	0	0	NA	32	0	0	0	
Withdrawals due to adverse events											
RCTs/Pts	0	8/1,563	9/1,266	5/1,161	6/1,867	3/769	8/1,346	4/1,192	3/1,105	48/10,965	
Frequency (%): AED/placebo	NA	9.6/4.5	13.3/3.4	5.6/2.7	15.8/5.8	13.0/5.8	6.5/5.2	13.1/5.1	14.4/5.2	11.8/4.7	
OR ^a	NA	2.34 (1.13–4.84)	3.23 (1.82–5.72)	2.31 (1.19–4.47)	2.96 ^b (2.00–4.37)	2.57 (0.98–6.71)	1.08 (0.58–1.99)	2.29 (1.32–3.99)	2.97 (1.79–4.94)	2.32 (1.90–2.82)	
I ² , %	NA	51	0	0	0	60	23	60	0	17	
Adverse effects											
Ataxia											
RCTs/Pts	2/961	12/1,524	7/918	2/578	5/1,689	1/297	4/904	2/647	3/1,105	40/9,122	
OR ^a	4.01 (2.28–7.07)	3.87 (2.51–5.96)	2.07 (1.21–3.52)	2.21 (1.19–4.13)	4.23 (2.49–7.21)	1.25 (0.44–3.59)	1.37 (0.45–4.21)	2.24 (0.92–5.45)	3.45 (1.44–8.26)	3.00 (2.44–3.69)	
I ² , %	0	0	0	0	0	NA	0	0	0	0	
Dizziness											
RCTs/Pts	2/961	13/1,763	9/1,229	6/1,206	6/1,867	4/850	9/1,697	4/1,192	3/1,105	58/12,639	
OR ^a	3.77 (2.51–5.65)	2.26 (1.49–3.42)	1.69 (1.19–2.39)	2.63 (1.68–4.09)	3.94 (2.90–5.35)	2.10 (1.28–3.45)	1.68 (0.92–3.05)	3.76 (2.32–6.09)	3.82 (2.31–6.33)	2.59 ^b (2.21–3.02)	
I ² , %	0	47	0	0	0	46	36	0	31	33	
Fatigue											
RCTs/Pts	2/961	12/1,547	8/1,092	4/952	6/1,867	3/769	5/1,183	2/647	3/1,105	47/10,622	
OR ^a	1.92 (1.17–3.15)	0.80 (0.55–1.18)	2.74 (1.81–4.14)	2.17 (1.18–3.99)	1.30 (0.94–1.80)	1.46 (0.53–4.02)	1.74 (1.16–2.61)	3.84 (1.50–9.86)	1.23 (0.53–2.88)	1.54 ^b (1.29–1.84)	
I ² , %	0	7	0	0	0	66	0	0	58	30	
Nausea											
RCTs/Pts	2/961	12/1,482	8/1,048	4/1,034	3/942	4/850	5/1,143	1/395	3/1,105	44/9,729	
OR ^a	3.71 (2.29–6.02)	1.96 (1.37–2.80)	2.42 (1.46–4.00)	0.93 (0.56–1.54)	1.11 (0.58–2.12)	1.45 (0.85–2.48)	0.81 (0.44–1.51)	1.09 (0.39–3.05)	2.34 (1.11–4.93)	1.65 ^b (1.37–1.99)	
I ² , %	0	0	0	0	0	0	0	NA	36	17	
Headache											
RCTs/Pts	2/961	1/281	8/966	6/1,206	6/1,867	2/554	7/1,358	4/1,192	3/1,105	39/9,644	
OR ^a	1.55 (0.88–2.73)	1.17 (0.66–2.08)	0.84 (0.56–1.25)	0.94 (0.50–1.76)	0.72 (0.46–1.12)	0.94 (0.51–1.74)	0.75 (0.41–1.36)	1.30 (0.76–2.23)	1.50 (0.98–2.29)	0.98 (0.81–1.17)	
I ² , %	62	1/281	19	32	46	0	48	18	0	39	
Somnolence											
RCTs/Pts	2/961	13/1,763	10/1,315	6/1,206	6/1,867	4/850	6/1,697	4/1,192	3/1,105	59/12,725	
OR ^a	2.94 (2.00–4.34)	1.44 (0.97–2.12)	2.43 (1.70–3.45)	2.34 (1.67–3.28)	2.32 ^b (1.70–3.18)	2.36 (1.18–4.72)	1.66 (1.17–2.35)	1.56 (0.86–2.83)	1.57 (0.89–2.76)	1.92 ^b (1.69–2.20)	
I ² , %	0	19	0	0	0	51	16	24	0	15	

Values in brackets represent the 95% confidence interval.
OXC, oxcarbazepine; LTG, lamotrigine; TPM, topiramate; GBP, gabapentin; PGB, pregabalin; ZNS, zonisamide; TGB, tiagabine; LEV, levetiracetam; EBZ, eslicarbazepine; LCS, lacosamide; AED, antiepileptic drug;
RCTs, randomized controlled trials; Pts, patients; OR, odds ratio; NA, not applicable.
^aOR estimate based on the random-effect model.
^bSignificant differences (p < 0.05) were found in the subgroup analysis for different daily doses. In the case of All AEDs, the subgroups represent each one of the different AED.
^cI² ≥ 50% indicates the presence of significant heterogeneity.

Values in brackets represent the 95% confidence interval.

OXC, oxcarbazepine; LTG, lamotrigine; TPM, topiramate; GBP, gabapentin; PGB, pregabalin; ZNS, zonisamide; TGB, tiagabine; LEV, levetiracetam; EBZ, eslicarbazepine; LCS, lacosamide; AED, antiepileptic drug; RCTs, randomized controlled trials; Pts, patients; OR, odds ratio; NA, not applicable.

^aOR estimate based on the random-effect model.^bSignificant differences ($p < 0.05$) were found in the subgroup analysis for different daily doses. In the case of All AEDs, the subgroups represent each one of the different AED.^c $\chi^2 \geq 50\%$ indicates the presence of significant heterogeneity.



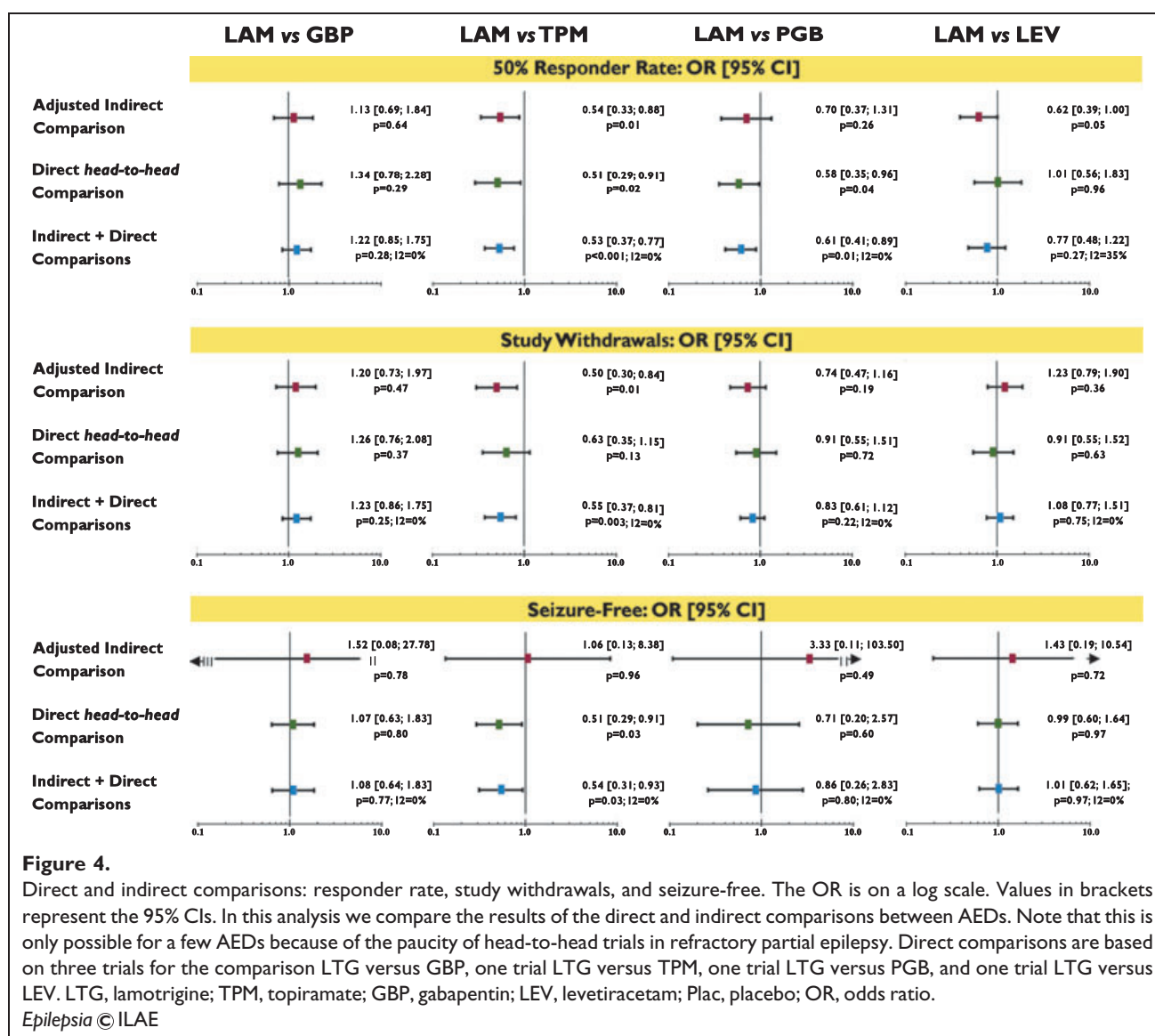
significantly less with LEV in comparison to all other AEDs. There were no differences between all the other AEDs.

Comparisons between the AEDs for the six prespecified adverse events showed few differences. There were no significant differences between AEDs for ataxia. Headache was more frequent with LCS (OR 1.61; CI 95% 1.01–2.57); dizziness was more frequent with PGB (OR 1.64; CI 95% 1.16–2.33); fatigue was more frequent with TPM (OR 1.96; CI 95% 1.25–3.07) and less frequent with LTG (OR 0.47; CI 95% 0.31–0.71); nausea was more frequent with OXC (OR 2.49; CI 95% 1.49–4.17) and less frequent with GBP

(OR 0.53; CI 95% 0.31–0.90) and LEV (OR 0.46; CI 95% 0.23–0.88); and somnolence was more frequent with OXC (OR 1.59; CI 95% 1.05–2.39) and less frequent with TGB (OR 0.60; CI 95% 0.39–0.92). Figure S2 shows the detailed results for pre-specified adverse events.

Combined evidence from indirect and direct comparisons

We included eight head-to-head trials; seven comparing LTG with other AEDs (three vs. GBP, one vs. TPM, one vs. PGB, one vs. LEV, and one vs. TGB) and one comparing TGB with TPM. The data available allowed us to combine



direct and indirect evidence for the primary and seizure-free outcomes for the comparisons LTG versus GBP, LTG versus TPM, LTG versus PGB, and LTG versus LEV (Fig. 4).

LTG versus GBP. Combined results for indirect and direct comparisons showed no differences in responder rate (OR 1.22; CI 95% 0.85–1.75; $I^2 = 0\%$), withdrawal rate (OR 1.23; CI 95% 0.86–1.75; $I^2 = 0\%$), or seizure-free rate (OR 1.08; CI 95% 0.64–1.83; $I^2 = 0\%$). There was no discrepancy between direct and indirect estimates ($p \geq 0.64$ for all outcomes).

LTG versus TPM. Combined analyses favored TPM for responder rate (OR 0.53; CI 95% 0.37–0.77; $I^2 = 0\%$) and seizure-free rate (OR 0.54; CI 95% 0.31–0.93; $I^2 = 0\%$), and LTG for withdrawal rate (OR 0.55; CI 95% 0.37–0.81;

$I^2 = 0\%$). There was no discrepancy between direct and indirect estimates ($p \geq 0.51$ for all outcomes).

LTG versus PGB. Combined analyses favored PGB for responder rate (OR 0.61; CI 95% 0.41–0.89; $I^2 = 0\%$). There were no differences in seizure-free rate (OR 0.86; CI 95% 0.26–2.83; $I^2 = 0\%$) or withdrawal rate (OR 0.83; CI 95% 0.61–1.12; $I^2 = 0\%$). There was no discrepancy between direct and indirect estimates ($p \geq 0.41$ for all outcomes).

LTG versus LEV. Combined results showed no significant differences in responder rate (OR 0.77; CI 95% 0.48–1.22; $I^2 = 35\%$), seizure-free rate (OR 1.01; CI 95% 0.62–1.65; $I^2 = 0\%$), or withdrawal rate (OR 1.08; CI 95% 0.77–1.51; $I^2 = 0\%$). There was no discrepancy between direct and indirect estimates ($p \geq 0.21$ for all outcomes).

DISCUSSION

Taken together, the most relevant findings of our study are: (1) TPM and probably LEV are more efficacious in controlling seizure frequency (as measured by analysis of the responder rate), whereas GBP is less efficacious in comparison to all other new AEDs; (2) tolerability (as measured by study withdrawals) was poorer with OXC and TPM, whereas GBP and LEV are better tolerated; and (3) globally, the frequency of the most common side effects is comparable between the new AEDs.

Although the combined results from direct and indirect comparisons were in close agreement, to date, only few head-to-head RCTs have been conducted in partial refractory epilepsy. Therefore, our conclusions are essentially based on indirect evidence. Indirect comparisons are being used increasingly for the evaluation of a wide range of healthcare interventions. However, because of its limitations, the potential usefulness of indirect comparisons is still overshadowed by concern about bias resulting from its misuse (Ioannidis, 2006; Song et al., 2009). The essence of indirect comparisons is that the effect for the comparison of a given intervention versus other intervention(s) is estimated using a common comparator (in our case the placebo). Therefore, adjusted indirect comparisons may counterbalance bias in trials and provide an unbiased estimate, if the two sets of trials are similarly biased. The consequence is that its validity depends on the internal validity and similarity of the included trials, especially the common comparator. Other limitations are: indirect comparisons are not randomized comparisons; it represents essentially observational findings across trials and may suffer the biases of observational studies, for example confounding; and the variance of comparison is always greater than that from direct comparison (wider 95% CI) and, therefore, the precision is smaller.

In view of these potential limitations, at least three assumptions must be fulfilled for the results from indirect comparisons be considered valid (Song et al., 2003, 2009).

First, trials should be sufficiently homogeneous, both clinically and methodologically, to be pooled together (homogeneity assumption). In our study, all trials enrolled patients with similar clinical characteristics and study designs (adequate allocation concealment, blinding of interventions, length of baseline and treatment periods, and outcomes measures). Similarity also existed across studies regarding the dose regimens of the concurrent and the new AEDs. In most placebo-controlled trials (67%; 42 of 63), the dose of concurrent AEDs were kept constant throughout the study and in an additional 16% of the trials the dose of existing AEDs could be modified to achieve a predetermined plasma concentration. This information is unclear or not available in 16% of the trials. The new AED being tested was administered at a predetermined fixed dosage regimen in 52% of the placebo-controlled studies (33 of 63), and in

38% (24 of 63) of the trials there was also a target prefixed dose, although it could be adjusted (reduced) if the new AED was not tolerated. In 6% (4 of 63) of the studies, the dose regimen of the new AED was adjusted to achieve a predetermined plasma concentration. The new AED doses were taken to the patients' limits of tolerance in only one study.

Although 80% of the studies enrolled patients who have previously failed 1–3 AEDs, the possibility exists that outcome results may be different according to the number of previously failed AEDs. The absence of detailed outcomes data in earlier published trials for individual groups based on the number of previous AEDs proclaims further analysis on this topic. Another potential bias is the relatively variable length of the treatment period in included trials, with most studies (45%) following patients after baseline for 8–12 weeks. Studies with longer follow-up periods may be more sensitive for detecting changes in seizure frequency, given the propensity of seizures to occur in clusters, as well as to show differences in withdrawal rates due to adverse effects. The similarity in the duration of the baseline period across studies (8 weeks) and the fact that almost all trials analyzed the data on an intention-to-treat basis reduces the potential of this bias. Furthermore, the lack of statistical or clinical heterogeneity in the results suggests that these potential biases have not compromised the results.

Five of the 63 placebo controlled trials (6.2%) were conducted in children, with a mean age across studies between 8 and 11 years. We included these trials because the effect size for all outcomes was not different from that found in adult trials.

Second assumption, trials should be similar for moderators of relative treatment effect (similarity assumption). In our study, global responder rate in the placebo control arm was 15.3%. Similar results were found for all individual AEDs, except for TGB (6.9%), EBZ (19.5%), and LCS (22.6%). The higher responder rate in the placebo group in EBZ and LCS trials were probably responsible for the lower OR estimates, suggesting that these AEDs are less efficacious. In fact, when comparisons were adjusted for the baseline risk there were no significant differences. The study design and the type of patients enrolled in EBZ and LCS trials were not apparently different from those of all other AEDs. Other potential factors that can influence the placebo response include the location and the study period (year) of the trials. Descriptive comparisons with other AED trials matched for location and study year do not suggest that this is the case in our analysis. These aspects should be further explored in a future work. One explanation may be that the proportion of patients that were already medicated with TPM and LEV (probably the most efficacious drugs) is higher than in other AEDs trials, and, therefore, the added benefit is of smaller magnitude. As in the case of responder rate analysis, the global rate of study withdrawals found in all AED trials (12.8%) was similar for all AEDs, except for OXC (19.9%) and TPM (6.9%). In the case of TPM, when

indirect comparisons were adjusted for the baseline risk, the differences were not significant.

The third assumption is the consistency needed to combine results of direct and indirect comparisons. In our case, there was no clinically meaningful heterogeneity or significant statistical discrepancy between the evidence from these different sources.

Even when all these assumptions are fulfilled, as in our case, one should keep in mind that indirect evidence is suitable for suggesting the presence or absence of clinically relevant differences between drugs, but results with a lower grade of uncertainty can be achieved only by head-to-head evidence.

For a given clinical indication where several competing intervention strategies exist, clinicians and decision makers of healthcare policy often have to choose the most effective treatment. This is clearly the case of partial refractory epilepsy for which the need for evidence about the relative effectiveness of competing interventions will be ever increasing. In the last 15 years, at least 15 new AEDs have been introduced. However, despite the current availability of these new AEDs, treatment outcomes remain poor, and approximately 30% of patients have uncontrolled disease (Perucca et al., 2007; Bialer & White, 2010). In fact, a recent systematic review by Beyenburg et al. (2010) about the efficacy of the new AEDs for refractory epilepsy concluded that its efficacy was disappointingly small, with absolute risk differences, in comparison to placebo, of 21% (CI 95% 19–24%; $I^2 = 57\%$) and 6% (CI 95% 4–8%; $I^2 = 74\%$) for the outcomes responder rate and seizure-free, respectively. We have focused on partial refractory epilepsy and in selected AEDs for which we were able to include more studies than Beyenburg and colleagues. Our efficacy results are even more disappointing [19% (CI 95% 17–21%; $I^2 = 59\%$) and 2% (CI 95% 2–3%; $I^2 = 43\%$) for the outcomes responder rate and seizure-free, respectively].

In our study, we decided to compare each individual AED against all other AEDs, and not to perform mixed indirect comparisons between individual AEDs, because this approach is probably closer to the real world clinical management of these patients and, therefore, more generalizable to patients in general. At least in high-income countries, most, if not all, patients with refractory partial epilepsy are medicated with several, usually two to three, AEDs (Schuele & Lüders, 2008; French & Faught, 2009; Kwan & Sperling, 2009). Furthermore, we used in our study a classic frequentist method that has the advantages of transparency and being easier to use than Bayesian methods.

The transferability of the findings from RCTs in refractory epilepsy to clinical practice is not straightforward. Therefore, it is important to evaluate how the results from combined direct and indirect estimates compare with those arising from observational studies, which frequently can be easily extrapolated to everyday clinical practice (Zaccara et al., 2006). The most practical and clinically meaningful

information that can be obtained from observational studies is the assessment of AED treatment retention rates, which reflect the combination of all possible reasons for medication termination, including ineffectiveness and intolerance, and also capture each individual patient's preference (Chung et al., 2007).

The results of our study are in close agreement with those from several large observational studies that followed patients with refractory epilepsy in tertiary epilepsy centers. According to these studies, tolerability of TPM is poor, with retention rates of only 30% at 4 years, mostly because of adverse events that accounted for about 65% of the discontinuations (Bootsma et al., 2004). Low-dose and slow-dose escalation of TPM are associated with a higher long-term tolerability (Cho et al., 2009). Long-term retention rates are also poor with the other AEDs. However, with the exception of OXC (Seneviratne et al., 2008), the main reason for patients with refractory epilepsy to discontinue treatment is related to perceived lack of efficacy and not to adverse effects. For example, in the cases of GBP and LTG, less than one-fourth of patients will still be taking these drugs after 6 years (Wong, 1997; Wong et al., 1999), although more recent studies report a higher retention rates (56% for LTG at 4 years) (Bootsma et al., 2008a). Nevertheless, treatment withdrawal due to perceived lack of efficacy is about twice more frequent in patients on GBP and LTG than in patients treated with TPM (Lhatoo et al., 2000). Retention rates with LEV are also poor (46–58% at 2–3 years) (Depondt et al., 2006; Bootsma et al., 2008b). However, in comparison with TPM, the retention rate appears to be higher with LEV (46 vs. 38% at 2 years), because LEV had a more favorable side-effect profile than TPM with comparable efficacy (Bootsma et al., 2008b). The results from the available observational studies of PGB and ZNS in patients with refractory epilepsy are also unsatisfactory, with retention rates of 40–60% at 1–2 years for PGB (Carreño et al., 2007; Brandt et al., 2009; Valentin et al., 2009), and 45% and 29% at 2 and 3 years (respectively) for ZNS (Wroe et al., 2008). For both AEDs, treatment discontinuation was mostly because of inefficacy.

Zaccara et al. (2006) performed a systematic review that included 10 retrospective and 14 prospective open-label studies in drug-resistant epileptic adult patients in whom a new AED was added to a previous treatment. Retention rate after 1 year was 20–25% for GBP, 40–60% for TPM and LTG, and 60–75% for LEV. According to the authors, the best efficacy profile (seizure-free for 6 months) was found for LEV followed by TPM, LTG, and GBP. The best tolerability profile (lower withdrawing due to adverse effects) was found for LEV and LTG, whereas TPM was the least well-tolerated AED. Taken together, our results are in agreement with these findings. However, we should keep in mind that there are other aspects of global tolerability and safety of AEDs that are not captured in these studies. For example, recently, a large cohort study suggests that the risk

of suicidal acts or violent deaths is significantly lower with TPM than with other AEDs (Patormo et al., 2010).

In the present era of evidence-based decision making, the highest quality research evidence is considered to be that emerging from head-to-head RCTs (Song et al., 2009). However, with few exceptions, the clinical development program of the new AEDs is designed to obtain regulatory approval for add-on indication on partial refractory epilepsy. From the regulatory perspective, this does not require an active comparator and, therefore, most of the trials are placebo-controlled. The consequence is that, at present, the best estimates of the clinical comparability of the new AEDs came from adjusted indirect comparisons. In fact, the limited amount and the quality of information available for the new AEDs is probably one of the reasons that help explain some discrepancies found in clinical guidelines and between these and other recommendations (Beghi, 2004).

The true effectiveness of a given drug is related to its benefits (efficacy, quality of life, level of satisfaction) and risks (including different aspects of safety) in the real world and in a defined population (Alexander & Stafford, 2009). Furthermore, it has to take into consideration local jurisdictional issues and cost-effectiveness as an additional criterion (Clement et al., 2009). Therefore, the process of clinical decision making is far from being based solely on the interpretation of the evidence provided by RCTs or meta-analysis. In our study, we evaluated only three aspects of the effectiveness of AEDs: efficacy in controlling seizures frequency, tolerability, and global side effects—all in a controlled clinical setting of randomized trials. Although there were differences between the new AEDs for these outcomes, these were of relatively small magnitude to allow a definitive conclusion about which one(s) has a superior effectiveness over the others. However, one possible approach to resume the combined efficacy and tolerability data found in our study is to define an arbitrary criterion for what can be considered a clinically relevant difference between a new AED and all other AEDs. Assuming that a 5% difference is clinically relevant, the AEDs that showed a more favorable global effectiveness (combined responder and withdrawal rates) were LEV (13%) and TPM (9%), whereas TGB (−10%), GBP (−7%), and LCS (−7%) had a less favorable effectiveness in comparison to other AEDs. The other AEDs were within these limits (PGB: 3%; EBZ: −3%; ZNS: −3%, OXC: −5%; and LTG: −5%).

We should keep in mind that the worldwide social burden of epilepsy is huge, directly affecting an estimated 50 million people, and that 80% of this burden is borne by the resource-poor countries. In these countries, many epileptic patients are not treated because of the cost of treatment, the lack of availability of AED, and sociocultural aspects (Radhakrishnan, 2009).

The main implication of our study may be that, in the year of 2011, the process of clinical decision making about the relative effectiveness of the new AEDs in partial refractory

epilepsy at a local or country level probably depends more on other aspects than on the available controlled randomized evidence. To change this scenario of uncertainty, there is an urgent need of direct randomized evidence as well as prospective data from representative cohort studies. It is highly unlikely that the pharmaceutical industry will sponsor this clinical research, because it can pose considerable financial risks to companies. Independent scientific societies interested in epilepsy must join efforts to find funding for conducting this research.

DISCLOSURE

This was an academic project not funded by any government or nongovernment grant. The authors declare no conflicts of interest. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

AUTHOR CONTRIBUTIONS

JC contributed to the concept and design, data acquisition, data analysis, and interpretation of the data; wrote the first draft of the manuscript; critically revised the manuscript; and gave final approval of the submitted manuscript. FF and RA contributed to data acquisition, data analysis, and interpretation of data; and gave final approval of the submitted manuscript. MB and AVC contributed to the concept and design and interpretation of data; critically revised the manuscript; and gave final approval of the submitted manuscript. CS contributed to interpretation of data, critically revised the manuscript, and gave final approval of the submitted manuscript.

REFERENCES

- Alexander GC, Stafford RS. (2009) Does comparative effectiveness have a comparative edge? *JAMA* 301:2488–2490.
- Beghi E. (2004) Efficacy and tolerability of the new antiepileptic drugs: comparison of two recent guidelines. *Lancet Neurol* 3:618–621.
- Beyenburg S, Stavem K, Schmidt D. (2010) Placebo-corrected efficacy of modern antiepileptic drugs for refractory epilepsy: systematic review and meta-analysis. *Epilepsia* 51:7–26.
- Bialer M, White HS. (2010) Key factors in the discovery and development of new antiepileptic drugs. *Nat Rev Drug Discov* 9:68–82.
- Bootsma HP, Coolen F, Aldenkamp AP, Arends J, Diepman L, Hulsman J, Lambrechts D, Leenen L, Majoie M, Schellekens A, de Krom M. (2004) Topiramate in clinical practice: long-term experience in patients with refractory epilepsy referred to a tertiary epilepsy center. *Epilepsy Behav* 5:380–387.
- Bootsma HP, Vos AM, Hulsman J, Lambrechts D, Leenen L, Majoie M, Savelkoul M, Schellekens A, Aldenkamp AP. (2008a) Lamotrigine in clinical practice: long-term experience in patients with refractory epilepsy referred to a tertiary epilepsy center. *Epilepsy Behav* 12:262–268.
- Bootsma HP, Ricker L, Diepman L, Gehring J, Hulsman J, Lambrechts D, Leenen L, Majoie M, Schellekens A, de Krom M, Aldenkamp AP. (2008b) Long-term effects of levetiracetam and topiramate in clinical practice: a head-to-head comparison. *Seizure* 17:19–26.
- Brandt C, May TW, Pohlmann-Eden B, Nieder E, Elsner H, Witte-Boelt K, Schuermann I, Ebner A. (2009) Retention rate of pregabalin in drug-resistant epilepsy: 1-year follow-up, single-centre observation in 105 consecutive, adult patients. *Seizure* 18:634–638.
- Bucher HC, Guyatt GH, Griffith LE, Walter SD. (1997) The results of direct and indirect treatment comparisons in meta-analysis of randomized controlled trials. *J Clin Epidemiol* 50:683–691.
- Carreño M, Maestro I, Molins A, Donaire A, Falip M, Becerra JL, Castillo J. (2007) Pregabalin as add-on therapy for refractory partial seizures in every day clinical practice. *Seizure* 16:709–712.

- Castillo S, Schmidt DB, White S. (2000) Oxcarbazepine add-on for drug-resistant partial epilepsy. *Cochrane Database Syst Rev* (3):CD002028.
- Chadwick DW, Marson AG. (2005) Zonisamide add-on for drug-resistant partial epilepsy. *Cochrane Database Syst Rev* (4):CD001416.
- Chaisewikul R, Privitera MD, Hutton JL, Marson AG. (2001) Levetiracetam add-on for drug-resistant localization related (partial) epilepsy. *Cochrane Database Syst Rev* (1):CD001901.
- Cho YJ, Heo K, Kim WJ, Jang SH, Jung YH, Ye BS, Song DB, Lee BI. (2009) Long-term efficacy and tolerability of topiramate as add-on therapy in refractory partial epilepsy: an observational study. *Epilepsia* 50:1910–1919.
- Chung S, Wang N, Hank N. (2007) Comparative retention rates and long-term tolerability of new antiepileptic drugs. *Seizure* 16:296–304.
- Clement FM, Harris A, Li JJ, Yong K, Lee KM, Manns BJ. (2009) Using effectiveness and cost-effectiveness to make drug coverage decisions: a comparison of Britain, Australia, and Canada. *JAMA* 302:1437–1443.
- Deeks JJ, Altman DG, Bradburn MJ. (2001) Statistical methods for examining heterogeneity and combining results from several studies in meta-analysis. In Egger M, Davey Smith G, Altman DG (Eds) *Systematic reviews in health care: meta-analysis in context*, 2nd ed. BMJ Publication Group, London, pp. 313–335.
- Depondt C, Yuen AW, Bell GS, Mitchell T, Koeppe MJ, Duncan JS, Sander JW. (2006) The long term retention of levetiracetam in a large cohort of patients with epilepsy. *J Neurol Neurosurg Psychiatry* 77:101–103.
- DerSimonian R, Laird N. (1986) Meta-analysis in clinical trials. *Control Clin Trials* 7:177–188.
- French JA, Faught E. (2009) Rational polytherapy. *Epilepsia* 50:63–68.
- Glenny AM, Altman DG, Song F, Sakarovich C, Deeks JJ, D'Amico R, Bradburn M, Eastwood AJ; International Stroke Trial Collaborative Group. (2005) Indirect comparisons of competing interventions. *Health Technol Assess* 9:1e144.
- Ioannidis JP. (2006) Indirect comparisons: the mesh and mess of clinical trials. *Lancet* 368:1470–1472.
- Jette N, Hemming K, Hutton JL, Marson AG. (2008) Topiramate add-on for drug-resistant partial epilepsy. *Cochrane Database Syst Rev* (3):CD001417.
- Kwan P, Sperling MR. (2009) Refractory seizures: try additional antiepileptic drugs (after two have failed) or go directly to early surgery evaluation? *Epilepsia* 50:57–62.
- Lhatoo SD, Wong IC, Polizzi G, Sander JW. (2000) Long-term retention rates of lamotrigine, gabapentin, and topiramate in chronic epilepsy. *Epilepsia* 41:1592–1596.
- Lozsadi D, Hemming K, Marson AG. (2008) Pregabalin add-on for drug-resistant partial epilepsy. *Cochrane Database Syst Rev* (1):CD005612.
- Marson AG, Kadir ZA, Hutton JL, Chadwick DW. (2000) Gabapentin add-on for drug-resistant partial epilepsy. *Cochrane Database Syst Rev* (3):CD001415.
- Paterno E, Bohn RL, Wahl PM, Avorn J, Patrick AR, Liu J, Schneeweiss S. (2010) Anticonvulsant medications and the risk of suicide, attempted suicide, or violent death. *JAMA* 303:1401–1409.
- Pereira J, Marson AG, Hutton JL. (2002) Tiagabine add-on for drug-resistant partial epilepsy. *Cochrane Database Syst Rev* (3):CD001908.
- Perucca E, French J, Bialer M. (2007) Development of new antiepileptic drugs: challenges, incentives, and recent advances. *Lancet Neurol* 6:793–804.
- Radhakrishnan K. (2009) Challenges in the management of epilepsy in resource-poor countries. *Nat Rev Neurol* 5:323–330.
- Ramaratnam S, Marson AG, Baker GA. (2001) Lamotrigine add-on for drug-resistant partial epilepsy. *Cochrane Database Syst Rev* (3):CD001909.
- Schuele SU, Lüders HO. (2008) Intractable epilepsy: management and therapeutic alternatives. *Lancet Neurol* 7:514–524.
- Seneviratne U, D'Souza W, Cook M. (2008) Long-term assessment of oxcarbazepine in a naturalistic setting: a retrospective study. *Acta Neurol Scand* 117:367–369.
- Song F, Altman DG, Glenny AM, Deeks JJ. (2003) Validity of indirect comparison for estimating efficacy of competing interventions: empirical evidence from published meta-analyses. *Br Med J* 326:472–475.
- Song F, Harvey I, Lilford R. (2008) Adjusted indirect comparison may be less biased than direct comparison for evaluating new pharmaceutical interventions. *J Clin Epidemiol* 61:455–463.
- Song F, Loke YK, Walsh T, Glenny AM, Eastwood AJ, Altman DG. (2009) Methodological problems in the use of indirect comparisons for evaluating healthcare interventions: survey of published systematic reviews. *Br Med J* 338:b1147.
- Valentin A, Moran N, Hadden R, Oakes A, Elwes R, Delamont R, Mullatti N, Nashef L. (2009) Pregabalin as adjunctive therapy for partial epilepsy: an audit study in 96 patients from the South East of England. *Seizure* 18:450–452.
- Wong ICK. (1997) New antiepileptic drugs: study suggests that a quarter of patients will still be taken the new drugs after six years. *Br Med J* 314:603–604.
- Wong IC, Chadwick DW, Fenwick PB, Mawer GE, Sander JW. (1999) The long-term use of gabapentin, lamotrigine, and vigabatrin in patients with chronic epilepsy. *Epilepsia* 40:1439–1445.
- Wroe SJ, Yeates AB, Marshall A. (2008) Long-term safety and efficacy of zonisamide in patients with refractory partial-onset epilepsy. *Acta Neurol Scand* 118:87–93.
- Zaccara G, Messori A, Cincotta M, Burchini G. (2006) Comparison of the efficacy and tolerability of new antiepileptic drugs: what can we learn from long-term studies? *Acta Neurol Scand* 114:157–168.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Data S1. Search strategy, study selection and data extraction.

Data S2. Statistical analyses.

Data S3. Study description.

Figure S1. Literature searches and results.

Figure S2. Indirect comparisons between each antiepileptic drug (AED) versus all other AEDs: most frequent adverse effects.

Table S1. Drug-resistant partial epilepsy studies: antiepileptic drug versus placebo (63 RCTs).

Table S2. Drug-resistant partial epilepsy studies: antiepileptic drug versus antiepileptic drug (8 RCTs).

Please note: Wiley-Blackwell is not responsible for the content or functionality of any supporting information supplied by the authors. Any queries (other than missing material) should be directed to the corresponding author for the article.